

Part G: Reconnaissance Report

Target Information

Website Name: GitHub

Website URL: <https://github.com>

Reason for Choosing the Target: GitHub is a globally recognized software development platform used by millions of developers and organizations. It is a publicly accessible website, making it an ideal target for learning passive reconnaissance techniques while following ethical hacking principles.

WHOIS Results

The WHOIS lookup provided important information about the GitHub domain, including the domain registration details, registrar information, registration and expiry dates, name servers, and domain status. This information helps identify who manages the domain and how it is maintained.

Findings:

- Domain Name: github.com
- Registrar: (Insert from your WHOIS screenshot)
- Registration Date: (Insert from screenshot)
- Expiry Date: (Insert from screenshot)
- Name Servers: (Insert from screenshot)
- Domain Status: (Insert from screenshot)

Observation: WHOIS information is useful for identifying domain ownership and administrative details while remaining within the scope of passive reconnaissance.

DNS Information

DNS enumeration revealed publicly available DNS records associated with GitHub.

The following records were identified:

- A Record
- MX Record
- NS Record
- TXT Record

Purpose of Each Record:

- **A Record:** Maps the domain name to its IPv4 address.
- **MX Record:** Identifies the mail servers responsible for email delivery.
- **NS Record:** Specifies the authoritative DNS servers for the domain.
- **TXT Record:** Stores text-based information such as domain verification and email security configurations.

Observation: These records provide valuable insight into the organization's network infrastructure and communication services.

Technologies Identified

Technology fingerprinting showed that GitHub uses several modern web technologies to deliver a secure and scalable platform.

Examples of technologies identified include:

- Web Server
- Content Delivery Network (CDN)
- JavaScript libraries/frameworks
- Security technologies
- Cloud-based infrastructure

Summary: GitHub uses a highly optimized technology stack designed to support millions of users worldwide while maintaining performance, availability, and security.

Security Headers

The HTTP security header analysis showed that GitHub implements multiple security headers to protect users from common web attacks.

The headers checked include:

- Content-Security-Policy (CSP)
- X-Frame-Options
- X-Content-Type-Options
- Strict-Transport-Security (HSTS)
- Referrer-Policy

Observation: These security headers help defend against threats such as cross-site scripting (XSS), clickjacking, MIME-type sniffing, and protocol downgrade attacks, improving the overall security of the website.

Robots.txt Findings

GitHub provides a publicly accessible **robots.txt** file that contains instructions for search engine crawlers.

The analysis revealed that:

- A robots.txt file is available.
- It contains rules controlling crawler access.
- Certain directories are excluded from indexing.
- The file assists search engines in efficiently crawling public content.

GitHub also provides a public sitemap to help search engines discover and index publicly accessible pages more effectively.

Overall Observations

The reconnaissance phase demonstrated how much useful information can be collected from publicly available sources without interacting directly with the target's internal systems. WHOIS records revealed domain management details, DNS records provided insight into the domain's infrastructure, technology fingerprinting identified key web technologies, HTTP security headers reflected the organization's security posture, and the robots.txt file showed crawler directives for search engines. All information was gathered using passive reconnaissance techniques, ensuring that no unauthorized access or exploitation occurred.

Conclusion (Approximately 170 Words)

The reconnaissance phase is one of the most important stages of ethical hacking because it allows security professionals to collect publicly available information about a target before conducting any authorized security testing. During this task, I learned how to use passive reconnaissance techniques to gather valuable information without interacting directly with the target's systems. By analyzing WHOIS records, DNS information, website technologies, HTTP security headers, and the robots.txt file, I gained a better understanding of GitHub's publicly exposed infrastructure and security practices. This exercise demonstrated that organizations unintentionally reveal a significant amount of technical information through publicly accessible resources, which can be valuable for both defenders and attackers. It also highlighted the importance of implementing strong security measures and carefully managing publicly available information. Most importantly, I learned that ethical hackers must always conduct reconnaissance responsibly, legally, and with proper authorization. Passive information gathering forms the foundation of an effective penetration test while ensuring compliance with ethical and professional standards.